

PRODUCT REQUIREMENTS DOCUMENT

PlayArena: Browser-Based Gaming Platform

Product Name	PlayArena
Version	1.0 Confidential
Game Catalog	50 to 100 HTML5 games across 8 categories
Personalisation	Favourites, recently played, coin wallet via localStorage (guests) / Supabase (registered users)
Recommended Stack	React · Vite · Tailwind CSS · Supabase Auth & Database

1. Product Overview

PlayArena is a modern browser-based gaming portal that enables visitors to instantly discover and play a curated catalog of HTML5 games without registration. The platform combines a guest-first experience with optional account creation, a virtual coin economy, daily rotating game features, achievement progression, and weekly leaderboard competitions to drive long-term engagement.

2. Objectives

- Enable frictionless, zero-registration gameplay for all visitors.
- Support optional account creation for cloud saves, cross-device sync, and leaderboard access.
- Deliver a virtual coin economy that rewards gameplay and drives game discovery.
- Provide personalised daily game recommendations and a rotating featured catalog.
- Offer administrators complete control over games, categories, rewards, and rotations.

3. Target Users

User Type	Description
Guest Players	Visitors who play instantly using locally stored progress.
Registered Users	Accounts with cloud saves, leaderboard access, and full dashboard.
Platform Administrators	Staff who manage games, categories, coins, and featured content.

4. Platform Structure

Area	Key Pages / Sections
Public Pages	Homepage · Browse Games · Individual Game pages · Category pages · Favourites · Recently Played · Login & Registration · Weekly Leaderboard
Guest Dashboard	Coin balance · Daily rewards · Recently played · Continue playing · Favourite games · Achievement badges
Registered User Dashboard	User profile · Coin wallet & transaction history · Daily rewards & streak · Personalised recommendations · Recently played · Achievement badge collection · Leaderboard ranking

Admin Dashboard	Secure login with session management · Game & category management · Coin reward configuration · Daily rotation management · User management · Platform analytics
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5. Feature Requirements

5.1 Game Catalog

The catalog must support 50 to 100 HTML5 games across 8 categories: Action, Arcade, Puzzle, Racing, Sports, Adventure, Strategy, and Casual. Additional categories may be added via the admin dashboard. All games must be open-source, properly licensed, and royalty-free.

Game Record Field	Details
Title & Thumbnail	Display name and 320×180 px cover image
Category	One of 8 preset categories (extensible via admin)
Short Description	Brief summary of gameplay
Coin Reward / Cost	Coins awarded on play; coins required to unlock premium content
Popularity Indicators	Play count and popularity score
Related Games	Up to 4 linked titles shown on the game detail page

5.2 Daily Rotation and Recommendations

- 25 to 30 games featured daily, rotating automatically from the full catalog.
- Rule-based recommendation engine tracking gameplay history, category preferences, and game popularity.
- Recommendations refresh daily and appear on the homepage and registered user dashboard.

5.3 Coin Economy

- Users earn coins by playing games, completing games, claiming daily rewards, and reaching achievement milestones.
- Coins are spent to unlock premium games and content sections.
- All coin transactions are recorded; balances are validated before deductions.
- Guest coin balances are stored in browser local storage; registered users have server-side coin wallets.

5.4 Achievement System

- Achievement types: gameplay milestones, daily activity, and category completion.
- Progress tracking and badge collection visible in both guest and registered dashboards.

5.5 Weekly Leaderboard

Attribute	Detail
Ranking Metric	Coins earned within the current week
Reset Frequency	Automatic reset every week
Display Fields	Username, rank, coins earned, games played
Recognition	Top performers receive milestone recognition at reset

5.6 Guest-to-Registered Sync

On account creation, guest progress stored in local storage - coin balance, favourites, recently played, and achievements - must be merged into the new account without data loss.

6. Technical Requirements

6.1 Recommended Stack

Layer	Technology
Frontend	React · Vite · Tailwind CSS
Backend & Auth	Supabase Authentication and Database
Guest Storage	Browser Local Storage

6.2 Non-Functional Requirements

Requirement	Target
Responsive Design	Desktop, tablet, and mobile viewports
Browser Support	Chrome, Firefox, Safari, Edge (latest 2 versions)
Game Embedding	Fullscreen HTML5 with fast load performance
Security	Secure authentication · validated coin transaction handling
SEO	SEO-friendly URLs and metadata on all public pages
Console Errors	Zero critical console errors during normal use
Codebase	Deployment-ready · clean and maintainable

7. Personalisation and SEO

Personalisation

- Recently Played: recorded on game start, max 20 titles, persisted across sessions, clearable by user.
- Favourites: heart icon on cards and detail pages, individual and bulk remove, stored in localStorage (guests) or Supabase (registered users).
- Continue Playing: last 3 to 4 interacted games shown on homepage and user dashboard.
- Coin Wallet: real-time balance displayed; transaction history available to registered users.

SEO

- SEO-friendly URLs (/game/slug, /category/slug), unique title and meta description per page.
- Open Graph tags on game detail and category pages.
- sitemap.xml covering all public pages; robots.txt blocking /admin; canonical tags on parameterised views.

8. Scope Restrictions (Version 1.0)

The following must not be built:

1. Real-money gaming, gambling mechanics, or payment gateway integration
2. Multiplayer gameplay, chat systems, or voice communication
3. Native mobile applications (iOS or Android)
4. Cryptocurrency, NFT, or blockchain features
5. Advanced AI recommendation models

- 6. Friend systems or social networking features
- 7. Multi-language support
- 8. Pop-up ads, interstitial ads, or newsletter forms
- 9. WordPress, Wix, or any third-party page builder
- 10. In-house game development

9. Client-Provided Assets

The following assets will be supplied by the client before development begins:

- Official PlayArena logo and branding assets in PNG (transparent background).
- The provided logo must be used consistently across the entire platform.
- Logo design, logo redesign, and branding creation are not in scope.

10. Acceptance Criteria

Delivery is accepted only when all of the following are confirmed:

- 11. Games load and play in-browser without errors on Chrome, Firefox, Safari, and Edge.
- 12. Guest users can play, earn coins, save favourites, and track recently played without registering.
- 13. Guest progress syncs correctly into a new account on registration without data loss.
- 14. Coins are awarded and deducted accurately with full transaction records.
- 15. Daily featured games rotate automatically from the catalog.
- 16. Favourites, recently played, and continue playing features function correctly.
- 17. Weekly leaderboard updates in real time and resets each week.
- 18. Admin dashboard enables full management of games, categories, coins, and rotations.
- 19. Responsive layouts function correctly across desktop, tablet, and mobile.
- 20. No critical console errors appear during normal platform use.
- 21. sitemap.xml covers all public pages and robots.txt blocks /admin.
- 22. README.md includes working setup and deploy instructions.
- 23. No horizontal scroll on any viewport from 320 px upward.
- 24. Fullscreen gameplay works on iOS Safari, Chrome Android, and desktop.

11. Deliverables

Deliverable	Location	Contents
Frontend Application	/Frontend/	React/Vite application
Backend Application	/Backend/	Supabase integration and API layer
Database Schema	/Database/	SQL schema and seed data
Documentation	/Documentation/	Installation, deployment, and admin guides
Assets	/Assets/	Icons, thumbnails, and category images